

Ian Laga

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Education

Ph.D. in Statistics

August 2022

Pennsylvania State University

Dissertation: *Everyone Counts: Advanced Methods for Estimating Marginalized Populations*

Advisors: Dr. Le Bao and Dr. Xiaoyue Niu

B.S. in Applied Mathematics

May 2017

University of Colorado – Boulder

Minor in Statistics and Computer Science

Professional Position

Assistant Professor of Statistics

August 2022 – Present

Montana State University

Bozeman, MT

Refereed Publications

[*PhD advisee]

9. Vogel*, B., Cummins, B., and **Laga, Ian** (2025). Accounting for correlation and censoring in Bayesian network scale-up method models. *Social Networks*, 84:101–109
doi: [10.1016/j.socnet.2025.07.005](https://doi.org/10.1016/j.socnet.2025.07.005)
8. **Laga, Ian**, Kunke, J. P., McCormick, T. H., and Niu, X. (2025). Estimating and correcting degree ratio bias in the network scale-up method. *Sociological Methods & Research*
doi: [10.1177/00491241251364233](https://doi.org/10.1177/00491241251364233)
7. Kunke, J. P., **Laga, Ian**, Niu, X., and McCormick, T. H. (2024). Comparing the robustness of simple network scale-up method estimators. *Sociological Methodology*, 54(2):385–403
doi: [10.1177/00811750241242791](https://doi.org/10.1177/00811750241242791)
6. **Laga, Ian**, Niu, X., Rucinski, K., Baral, S., Rao, A., Chen, D., Viswasam, N., Phaswana-Mafuya, N. R., Diouf, D., Sabin, K., et al. (2023b). Mapping the number of female sex workers in countries across Sub-Saharan Africa. *Proceedings of the National Academy of Sciences*, 120(2):e2200633120
doi: [10.1073/pnas.2200633120](https://doi.org/10.1073/pnas.2200633120)
5. **Laga, Ian**, Bao, L., and Niu, X. (2023a). A correlated network scale-up method: Finding the connection between subpopulations. *Journal of the American Statistical Association*, 118(543):1515–1524
doi: [10.1080/01621459.2023.2165929](https://doi.org/10.1080/01621459.2023.2165929)
4. **Laga, Ian**, Niu, X., and Bao, L. (2022). Modeling the marked presence-only data: A case study of estimating the female sex worker size in Malawi. *Journal of the American Statistical Association*, 117(537):27–37
doi: [10.1080/01621459.2021.1944873](https://doi.org/10.1080/01621459.2021.1944873)

3. **Laga, Ian**, Bao, L., and Niu, X. (2021). Thirty years of the network scale-up method. *Journal of the American Statistical Association*, 116(535):1548–1559
doi: [10.1080/01621459.2021.1935267](https://doi.org/10.1080/01621459.2021.1935267)
2. **Laga, Ian** and Niu, X. (2020). Review of *Model-Based Geostatistics for Global Public Health: Methods and Applications*: by Peter J. Diggle and Emanuele Giorgi. *Journal of the American Statistical Association*, 115(530):1030–1032
doi: [10.1080/01621459.2020.1759988](https://doi.org/10.1080/01621459.2020.1759988)
1. **Laga, Ian** and Kleiber, W. (2017). The modified Matérn process. *Stat*, 6(1):241–247
doi: [10.1002/sta4.152](https://doi.org/10.1002/sta4.152)

Submitted Manuscripts

[*PhD advisee]

6. Thiele*, J., FireMoon, P., Johnson, O., Reum, M., **Laga, Ian**, and Rink, E. (2025+a). Condom use self-efficacy with Indigenous youth: The integration of factor analysis with participatory research. *Minor revisions at AIDS Care*
5. Thiele*, J., FireMoon, P., Johnson, O., **Laga, Ian**, and Rink, E. (2025+c). “If I didn’t have that, then I really don’t know where I’d be”: Examining the association between depression and participation in traditional activities for Indigenous youth using a concurrent triangulation design. *Submitted to Social Science & Medicine*
4. Thiele*, J., FireMoon, P., Johnson, O., **Laga, Ian**, and Rink, E. (2025+b). The development and testing of a pilot intervention to address sexual risk behavior, substance use, and mental health among Indigenous youth ages 12–18 years old. *Submitted to Journal of Community Health*
3. Sanei, S., **Laga, Ian**, Weir, S., and Bao, L. (2025+). A case-control sampling strategy for zero-inflated models with an application to female sex worker mapping in Sub-Saharan Africa. *Major revisions at The Annals of Applied Statistics*
2. Ma, J., **Laga, Ian**, and Johnson, E. (2025+). What drives hospital reporting gaps? hidden evidence from a national dataset. *Submitted to The Journal of the American Medical Informatics Association*
1. Fluer, A., **Laga, Ian**, Graham, L., Almirol, E., Meyer, M., Schneider, J., and Cummins, B. (2025+). From survey data to social multiplex models: Incorporating interlayer correlation from multiple data sources. *Submitted to Social Networks*

Manuscripts In Preparation

2. **Laga, Ian**. Evaluating the generalized network scale-up method estimator: Evidence from migrant domestic workers in Jordan. *Manuscript in preparation for Journal of the Royal Statistical Society: Series C*
1. **Laga, Ian**. Survey-based estimation of probe group sizes in the network scale-up method: A case study from Jordan. *Manuscript in preparation for Sociological Methodology*

Software

2. **Ian Laga** and Boateng, F. (2024). *MCPModGeneral: A Supplement to the ‘DoseFinding’ Package for the General Case*. R package version 0.1-3. Available at [CRAN](https://cran.r-project.org/web/packages/MCPModGeneral/index.html)

1. **Ian Laga**, Bao, L., and Niu, X. (2024). *networkscaleup: Network Scale-Up Models for Aggregated Relational Data*. R package version 0.1-2. Available at [CRAN](#)

Funded Grants

4. Counteracting Structural Barriers to Increase Access to Medications for Opioid Use Disorder (MOUD) Among Unhoused Montanans
Period: 2023-2026
Amount: \$22,495.8
Granting Agency: Open Aid Alliance (OPEAID)
Role: Co-PI
Primary Goals: Design and implement a stepped-wedge design to evaluate the effect of MOUD care based in syringe service programs in Montana.
3. Integrating Bayesian Analysis and Community-Based Participatory Research to Understand the Association Between Sexual Risk Behavior, Depression, Substance Use, and Protective Factors (BCSP)
Period: 2025
Amount: \$2,000
Granting Agency: Ivan Doig Center, Montana State University
Role: Principal Investigator
Primary Goals: Study American Indian youth health using Community-Based Participatory Research
2. Statistical Methodology for Model Checking for Aggregated Relational Data
Period: 2025
Funding Amount: Lodging, meals, and travel expenses for two weeks at a BIRS site for all collaborators
Granting Agency: Pacific Institute for the Mathematical Sciences (PIMS) and the Banff International Research Station (BIRS)
Collaborators: Dr. Owen Ward (Simon Fraser University), Dr. Anna Smith (University of Kentucky), Jieyun Wang (University of Kentucky), Benjamin Vogel (Montana State University)
Primary Goals: Develop methods to check the validity of Aggregated Relational Data models
1. Statistical Models for Estimating and Projecting HIV/AIDS Epidemics
Period: 2022-2024
Amount: \$72,809.00
Granting Agency: Subaward from Pennsylvania State University from funding received via National Institute of Allergy and Infectious Diseases, Project Number: 5 R01 AI136664-05
Role: Principal Investigator for MSU portion
Primary Goals: Develop methods to estimate the size of HIV high-risk populations, such as female sex workers; present the results at conferences; prepare and submit manuscripts for publication to peer-reviewed journals

Grant Proposals Submitted (Under Review or Unsuccessful)

4. Psychological Trauma, Occupational Stress, and Suicide Risks among Middle-Level Nurse Managers: Pathways for Prevention
Period: Submitted 2025
Granting Agency: American Foundation for Suicide Prevention
Collaborators: Dr. Yitong Wang (Montana State University)
Status: Under review

3. Validating a Primary Care Screening Tool to Identify Immigration-Related Social Determinants of Health
Period: Submitted 2025
Granting Agency: US Department of Health and Human Services
Collaborators: Dr. Sally Moyce (Montana State University)
Role: Co-PI
Status: Unsuccessful
2. Counting the Undercounted: Enhancing the Network Scale-up Method for Estimating Hard-to-Reach Populations
Period: Submitted 2024
Granting Agency: National Science Foundation
Collaborators: Dr. Xiayoue Niu (Pennsylvania State University)
Role: Principal Investigator
Status: Unsuccessful (under revision)
1. Improved Function for Children with CP Using Hippotherapy, the Equine Environment
Period: Submitted 2023
Granting Agency: Montana INBRE
Collaborators: Dr. Julia Mazzarella (University of Montana) and Dr. Stephanie Dimitroff (University of Montana)
Role: Co-PI
Status: Unsuccessful

Invited Seminars and Talks

4. “The Network Scale-up Method and the Degree Ratio.” Simon Fraser University Statistics & Actuarial Science Seminar, Burnaby, BC, Canada, December 1, 2023
3. “A Correlated Network Scale-up Model: Finding the Connection Between Subpopulations.” ASA/IMS Spring Research Conference, Banff, AB, Canada, May 24, 2023
2. “A Case-Control Sampling Strategy for Zero-Inflated Models with an Application to Female Sex Worker Mapping in Sub-Saharan Africa.” Montana American Statistical Association Chapter Meeting, Bozeman, MT, USA, October 28, 2022
1. “A Correlated Network Scale-up Model: Finding the Connection Between Subpopulations.” Stochastic Modeling and Computational Statistics Talk, Department of Statistics, Pennsylvania State University, PA, USA, September 17, 2021

Contributed Seminars and Talks

5. “Studying Migrant Workers Using the Network Scale-up Method.” Joint Statistical Meetings, Nashville, TN, USA, August 4, 2025
4. “Correlated Models for Aggregated Relational Data and Social Network Sizes.” Joint Statistical Meetings, Portland, OR, USA, August 7, 2024
3. “The Analysis of Aggregated Relational Data Collected From Surveys With ‘How Many X’s Do You Know?’ Questions.” International Conference on Establishment Statistics, Glasgow, Scotland, UK, June 19, 2024

2. "Finding the Hidden Populations: A Correlated Network Scale-up Model." Joint Statistical Meetings, Virtual Conference, August 12, 2021
1. "Finding the Hidden Populations: A Correlated Network Scale-up Model." World Meeting of the International Society for Bayesian Analysis, Virtual Conference, June 16, 2021

Advising

All students are earning Statistics degrees unless otherwise noted

Current PhD Students

Ben Vogel	2024-Present
Jeffrey Thiele (Indigenous and Rural Health), co-advised with Dr. Beth Rink	2024-Present

Statistics MS Writing Project

Mark Braun	2024-2025
Bernard Ntiamoah	2024-2025
Jacob Oard	2023-2024
Michael Hessler	2023-2024
Gifty Osei	2022-2023
Michael Kumi	2022-2023

PhD Thesis Committee

Sarah Mensah	2024-Present
Steven Hammond	2024-Present
Korathotage Lakviru Perera	2024-Present
Clinton Patrick Pollock	2024-Present
Madeline Metcalf (Indigenous and Rural Health)	2024-Present

Teaching

Montana State University

STAT 532: Bayesian Data Analysis	Fall 2025
STAT 510: Statistical Consulting Seminar	Fall 2025, Spring 2025, Fall 2024
STAT 506: Advanced Regression Analysis	Spring 2025, Spring 2024, Spring 2023
STAT 448: Mixed Effects Models	Fall 2024, Fall 2022
STAT 437: Introduction to Applied Multivariate Analysis	Fall 2023
STAT 425: Biostatistical Data Analysis	Fall 2023
STAT 441/541: Experimental Design	Spring 2023

Pennsylvania State University

STAT 440: Computational Statistics	Spring 2020
STAT 415: Mathematical Statistics	Fall 2018

Service

Montana State University

Accelerated Master's Program	2024-Present
<i>Advisor</i>	
Statistics Undergraduate Program	2024-Present
<i>Advisor</i>	
Statistics Graduate Certificate	2024-Present
<i>Advisor</i>	
DataFest	2024
<i>Judge</i>	
Statistics Graduate Student Admissions Subcommittee	2023
<i>Member</i>	
Website Committee	2022-Present
<i>Member</i>	
Jeffrey Thiele, Department of Health and Human Development	2023-Present
<i>Co-mentor for TL1 grant</i>	

Professional Service

Montana Chapter of the American Statistical Association	2022-Present
<i>Treasurer</i>	
United States Geological Survey	2024-2025
<i>Professional consultation and collaboration to publish R code as a package on CRAN</i>	
United Nations: International Organization for Migration	2024-Present
<i>Professional consultation and collaboration to design and implement a Network Scale-up Method survey in Jordan</i>	

Peer Review

<i>PLOS Global Public Health</i>	2025
<i>Journal of the Royal Statistical Society: C</i>	2024, 2025
<i>Sociological Methodology</i>	2024, 2025
<i>Quality and Quantity</i>	2024
<i>Field Methods</i>	2023

Awards

2. **2020 Teaching Award Honorable Mention:** Pennsylvania State University Statistics Department
1. **2019 NSF Graduate Research Fellowship Honorable Mention:** National Science Foundation

Professional Memberships

American Statistical Association (ASA)